



Figure 3: 20/50 moving average crossover

```
ax.scatter(
    df.index[sell],
    df.loc[sell, "Close"],
    marker="v",
    s=120,
    label="SELL",
    zorder=5
)
ax.set_title(
    f"{ticker} - 20/50 Moving Average Crossover"
)
ax.set_xlabel("Date")
ax.set_ylabel("Price")
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

The chart now provides a visual representation of the positioning algorithm (Figure 3).

How the strategy helps to hold stock

Finally, for a clear demonstration of the Hold and Sell periods along the entire moving average cycle we can shade the holding periods of the stock. This is particularly useful for understanding the trading pattern.

```
holding = df["Signal"] == 1
```

The shading of the periods during which the strategy helps to hold the stock is implemented here and

shown in Figure 5:

```
fig, ax = plt.subplots(figsize=(15, 7))
ax.plot(
    df.index,
    df["Close"],
    label="Close Price",
    linewidth=1.2
)
ax.plot(
    df.index,
    df["SMA20"],
    label="20-Day SMA",
    linewidth=2
)
ax.plot(
    df.index,
```

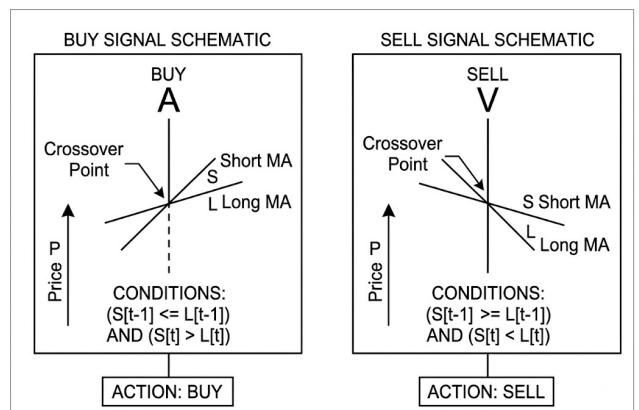


Figure 4: Buy signal and Sell signal schematics